Huan Li — RESUME

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Research interests: **databases**, **data-intensive systems**, **machine learning**, and **mobile computing**.

Education

Doctor's Degree — **Zhejiang University, China**
Computer Science and Software Engineering — *01/09/2012–30/06/2018*
Thesis: "*Uncertain Indoor Mobility Data Mining and Analytics*"; Advisors: Prof. Gang Chen and Prof. Lidan Shou.

Bachelor's Degree — **Sichuan University, China**
Computer Science — *01/09/2008–30/06/2012*
GPA score ranked 2nd among the 466 students.

Employment

Postdoctoral Research Fellow — **Aalborg University**
Department of Computer Science — *08/05/2019–Present*

Senior Algorithm Engineer — **Ant Financial, Alibaba Group**
LBS Data Intelligence Team — *15/07/2018–05/05/2019*

Research Assistant — **Aalborg University**
Department of Computer Science — *01/09/2014–01/02/2015*

Publications

- Journal Articles

- **Huan Li**, Hua Lu, Gang Chen, Ke Chen, Qinkuang Chen, Lidan Shou. Towards Translating Raw Indoor Positioning Data into Mobility Semantics. *ACM Transactions on Data Science* (TDS), 35 pages, 2020.
- **Huan Li**, Hua Lu, Lidan Shou, Gang Chen, Ke Chen. Finding Most Popular Indoor Semantic Locations Using Uncertain Mobility Data. *IEEE Trans. Knowl. Data Eng.*, 31(11): 2108-2123, 2019.
- **Huan Li**, Hua Lu, Lidan Shou, Gang Chen, Ke Chen. In Search of Indoor Dense Regions: An Approach Using Indoor Positioning Data. *IEEE Trans. Knowl. Data Eng.*, 30(8): 1481–1495, 2018.

- Conference Papers

- **Huan Li**, Hua Lu, Muhammad Aamir Cheema, Lidan Shou, Gang Chen. Indoor Mobility Semantics Annotation Using Coupled Conditional Markov Networks. *IEEE ICDE*, 12 pages, 2020.
- Tiantian Liu, Zijin Feng, **Huan Li**, Hua Lu, Muhammad Aamir Cheema, Hong Cheng, and Jianliang Xu. Shortest Path Queries for Indoor Venues with Temporal Variations. *IEEE ICDE*, 4 pages, 2020. (Short Paper)
- Zijin Feng, Tiantian Liu, **Huan Li**, Hua Lu, Lidan Shou, Jianliang Xu. Indoor Top-k Keyword-aware Routing Query. *IEEE ICDE*, 12 pages, 2020.
- **Huan Li**, Hua Lu, Lidan Shou, Gang Chen, Ke Chen. Finding Most Popular Indoor Semantic Locations Using Uncertain Mobility Data (extended abstract). *IEEE ICDE*, 2139-2140, 2019.
- **Huan Li**, Hua Lu, Lidan Shou, Gang Chen, Ke Chen. In Search of Indoor Dense Regions: An Approach Using Indoor Positioning Data (extended abstract). *IEEE ICDE*, 2127–2128, 2019.
- **Huan Li**, Pai Peng, Hua Lu, Lidan Shou, Ke Chen, Gang Chen. E^2C^2: Efficient and Effective Camera Calibration in Indoor Environments. *ACM UbiComp*, 9–12, 2015. (Poster)

- Demonstrations

- **Huan Li**, Feichao Shi, Hua Lu, Gang Chen, Ke Chen, Lidan Shou. TRIPS: A System for Translating Raw Indoor Positioning Data into Visual Mobility Semantics. *PVLDB*, 11(12):1918–1921, 2018.
- **Huan Li**, Hua Lu, Xin Chen, Gang Chen, Ke Chen, Lidan Shou. Vita: A Versatile Toolkit for Generating Indoor Mobility Data for Real-World Buildings. *PVLDB*, 9(13):1453–1456, 2016.

- Patents

- Lidan Shou, **Huan Li**, Ke Chen, Xinyuan Luo, Gang Chen, Dawei Jiang, Tianlei Hu. Indoor Semantic Trajectory Labeling and Complementing Method in Low-sampling Positioning Environment. CN201810778020.X, published on 11 Jan, 2019.
- Ke Chen, **Huan Li**, Lidan Shou, Xinyuan Luo, Gang Chen, Dawei Jiang, Sai Wu. An Approximate Method for Searching Check-in Users with Non-real-time Positioning Data. CN201810778007.4, published on 30 Nov, 2018.
- Gang Chen, **Huan Li**, Lidan Shou, Ke Chen, Xinyuan Luo, Sai Wu, Tianlei Hu. Indoor Continuous Flow Monitoring Method for Probabilistic Streaming Mobility Data. CN201810779777.0, published on 30 Nov, 2018.

Teaching

17/01/2020–present: Supervisor of SW8 project (Spring 2020) at Aalborg University.

01/09/2019–present: Co-supervising R.A. Bang Van Bui in the topic of Wi-Fi positioning data cleansing.

01/09/2019–20/01/2020: Supervisor of SW5 project (Autumn 2019) at Aalborg University.

01/05/2019–present: Co-supervising Ph.D. student Tiantian Liu in the topic of context-aware queries for indoor LBS.

01/03/2014–30/06/2014: Teaching assistant for "Operating System" (Spring 2014) at Zhejiang University.

01/09/2013–30/01/2014: Teaching assistant for "Advanced Database Technology" (Autumn 2013) at Zhejiang University.

Professional Services

✶ **PC member** for SIGSPATIAL ISA 2018, and MDM ALIAS 2019.

✶ **Journal referee** for TKDE, Journal of GIS, and Journal of Mobile Information Systems.

✶ **External reviewer** for VLDB, ICDE, UbiComp, WWW, CIKM, DASFAA, SIGSPATIAL, IEEE BigData, and ICMR.

Other Services

✻ **Volunteer leader** for VLDB 2014 and NDBC 2017.

✻ Campus Ambassador of Hulu Inc. at Zhejiang University from 2012 to 2015.

✻ Monitor of the No.1 PhD Class of 2012 of Zhejiang University from 2012 to 2018.

✻ Monitor of the College's Innovation Class of Sichuan University from 2009 to 2012.

Selected Awards

✯ Outstanding Student of 41st Guards Algorithm Class of Ant Financial (2018).

✯ Candidate for Outstanding Doctoral Thesis of Zhejiang University (2018).

✯ Alibaba Big Data Algorithm Contest (2016): Spatio-temporal distribution prediction of airport passengers (top-10/3455).

✯ Triple-A Graduate Award of Zhejiang University (2015).

✯ **Outstanding Ph.D. Student Scholarship of Zhejiang University** (2015/2016).

✯ Outstanding Graduate Leader Award of Zhejiang University (2014).

✯ Award of Honor for Graduate of Zhejiang University (2013/2015/2016).

✯ **Creative Talents Award of Sichuan University** (2012).

✯ **China National Scholarship** (2009/2010/2011).

Grants

✯ VLDB Travel Grant (2016).

✯ **EU Marie Skłodowska-Curie Individual Fellowships** (2020), 219,000 EUR.

Participated Projects

Data Management Foundations for Indoor LBS — **Aalborg University**
Independent Research Fund Denmark — *01/05/2019–Present*

iLocation: Platform for Location-based Service at Ant Financial — **Ant Financial (Alibaba)**
Project at LBS Data Intelligence Team — *15/07/2018–05/05/2019*

Retail Industry Data Analysis Platform — **Zhejiang University**
Project at Database Laboratory — *01/03/2015–31/12/2017*

Management and Mining of Complex Data in Intelligent Indoor Environment — **Zhejiang University**
NSFC Grant No. 61672455 — *01/07/2015–01/05/2016*

Multi-Source and Heterogeneous Trajectory Data Fusion — **Zhejiang University**
NSFC Grant No. 6161101183 — *01/03/2015–01/03/2017*

Management and Application of Indoor Trajectory Data. — **Aalborg University**
NSFC Grant No. 61528207 (Joint Research Fund for Overseas) — *01/09/2014–28/02/2018*

Human-Computer Fusion based Crowd Cognition: Theory and Models — **Zhejiang University**
973 Program of China, Grant No. 2015CB352402 — *01/12/2013–30/06/2015*

AR-based Mobile Social Application — **Sichuan University**
The Innovative Experimental Program of China — *01/09/2010–31/08/2011*